

Mathematics Reference: L.C.M. and H.C.F.

Educational Series

December 29, 2025

1 Introduction

Understanding the relationship between factors and multiples is essential for solving problems involving fractions, ratios, and algebraic expressions.

1.1 H.C.F. (Highest Common Factor)

The H.C.F. of two or more numbers is the greatest number that divides each of them exactly. It is also known as the **G.C.D. (Greatest Common Divisor)**.

1.2 L.C.M. (Least Common Multiple)

The L.C.M. of two or more numbers is the smallest number which is exactly divisible by each of them.

2 Fundamental Formulas

2.1 The Product Rule

For any two numbers a and b :

$$\text{Product of numbers} = \text{H.C.F.} \times \text{L.C.M.}$$

$$a \times b = \text{H.C.F.}(a, b) \times \text{L.C.M.}(a, b)$$

2.2 L.C.M. and H.C.F. of Fractions

When dealing with fractions, use the following rules:

1. **H.C.F. of Fractions** = $\frac{\text{H.C.F. of Numerators}}{\text{L.C.M. of Denominators}}$
2. **L.C.M. of Fractions** = $\frac{\text{L.C.M. of Numerators}}{\text{H.C.F. of Denominators}}$

3 Methods of Calculation

3.1 Prime Factorization Method

To find H.C.F. and L.C.M. of 12 and 18:

- $12 = 2^2 \times 3^1$

- $18 = 2^1 \times 3^2$

H.C.F.: Product of the lowest powers of common factors.

$$\text{H.C.F.} = 2^1 \times 3^1 = 6$$

L.C.M.: Product of the highest powers of all prime factors involved.

$$\text{L.C.M.} = 2^2 \times 3^2 = 36$$

4 Important Properties

- The H.C.F. of two co-prime numbers is always 1.
- The L.C.M. of two co-prime numbers is their product ($a \times b$).
- The H.C.F. of a group of numbers is always a factor of their L.C.M.
- H.C.F. is always less than or equal to the smallest number in the set.
- L.C.M. is always greater than or equal to the largest number in the set.

5 Practice Examples

Example 1: Find the H.C.F. and L.C.M. of $\frac{2}{3}, \frac{8}{9}, \frac{16}{81}$.

- $\text{H.C.F.} = \frac{\text{HCF}(2,8,16)}{\text{LCM}(3,9,81)} = \frac{2}{81}$

- $\text{L.C.M.} = \frac{\text{LCM}(2,8,16)}{\text{HCF}(3,9,81)} = \frac{16}{3}$